

Digital Signal Processing

Fourier Series and Transforms

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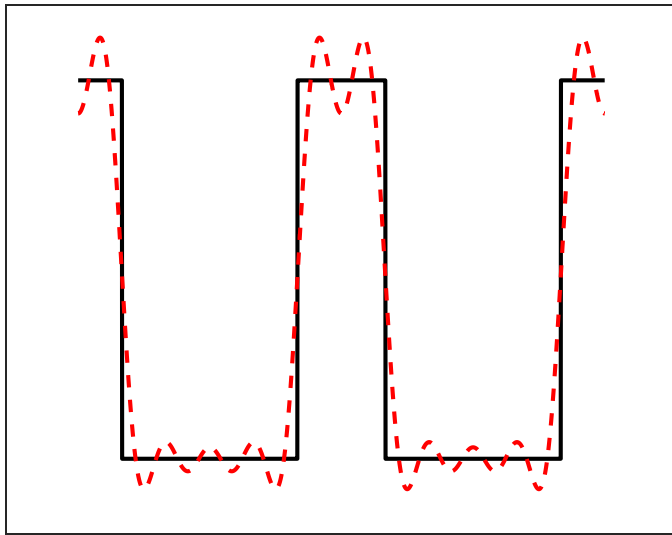
A signal can be viewed in time or frequency

The basis functions of *Fourier analysis tools* are sinusoids

Unlike the DFT, these sinusoids have infinite duration

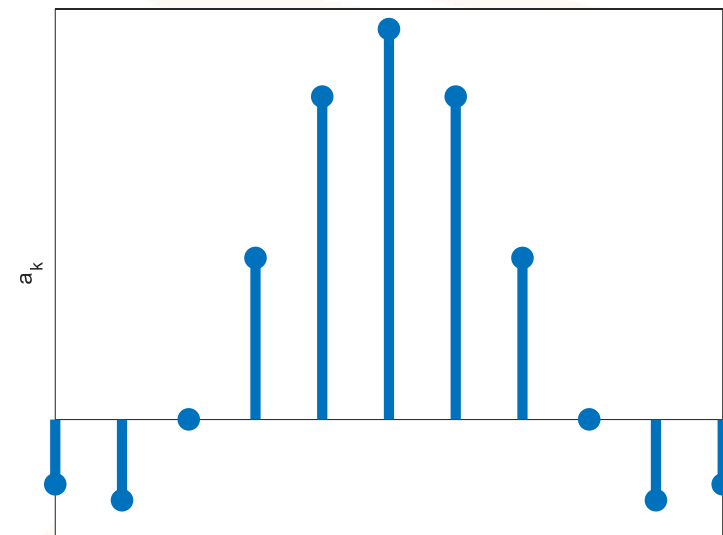
$$x(t) = x(t + T)$$

$$x(t) = \sum_k c_k \exp(jk2\pi t/T)$$



A

A^{-1}



Fourier Series are “a special case” of Fourier Transforms

	Continuous-Time	Discrete-Time
Periodic	Fourier Series (FS) V	Discrete-Time Fourier Series (DTFS) V
Non-periodic	Fourier Transform (FT) V/Hz	Discrete-Time Fourier Transform (DTFT) V/rad

Periodic

Continuous-Time

Fourier Series (FS)

Discrete-Time

Discrete-Time Fourier Series (DTFS)



Non-periodic

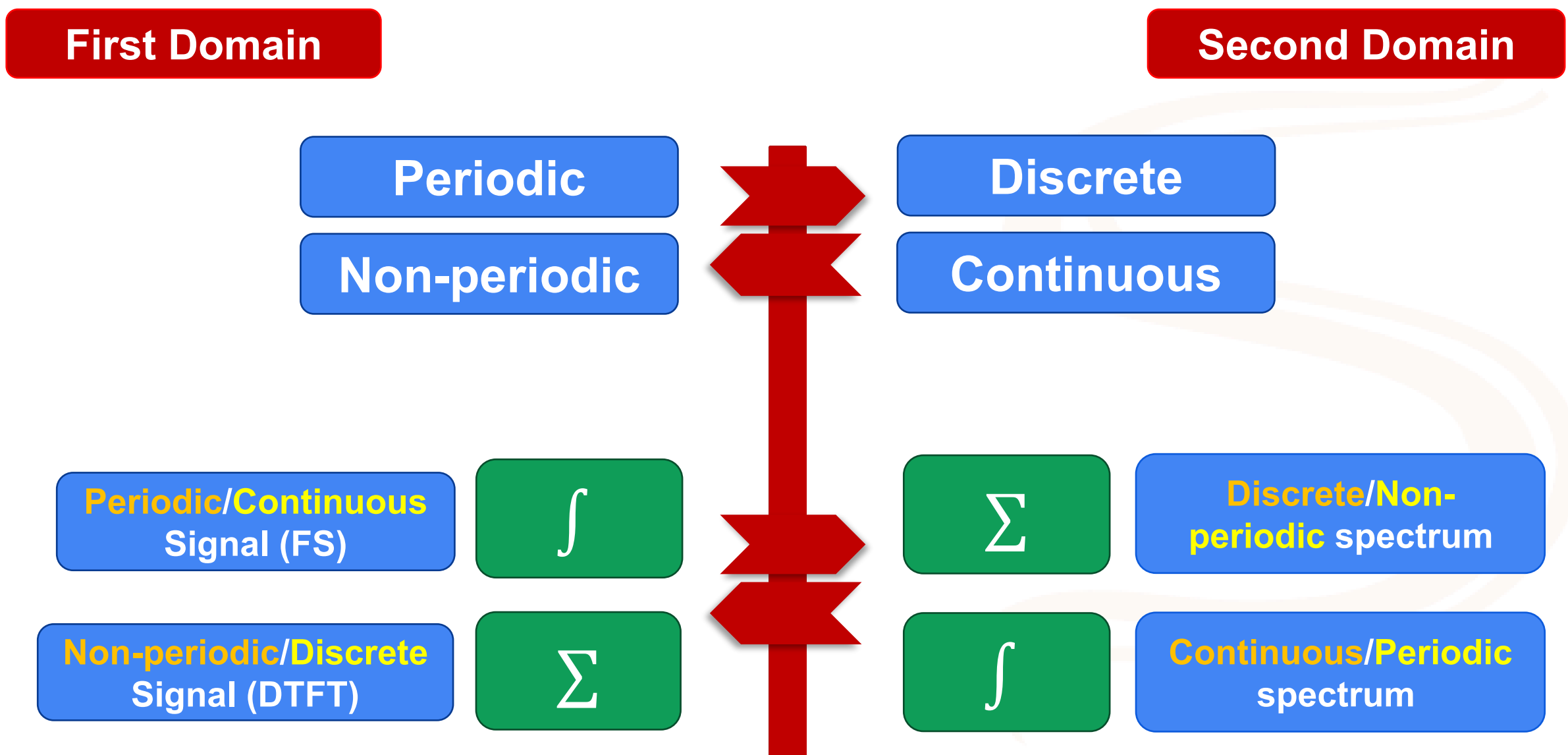
Fourier Transform (FT)

V/Hz

Discrete-Time Fourier Transform (DTFT)

V/rad

Duality between Time and Frequency domains



Duality between Time and Frequency domains

The big picture

	Continuous Time	Discrete Time
Fourier Series	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$ <u>continuous and periodic in time</u> $a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$ <u>discrete and aperiodic in frequency</u>	$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk \frac{2\pi}{N} n}$ <u>discrete and periodic in time</u> $a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n}$ <u>discrete and periodic in frequency</u>
Fourier Transform	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$ <u>continuous and aperiodic in time</u> $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$ <u>continuous and aperiodic in frequency</u>	$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$ <u>discrete and aperiodic in time</u> $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$ <u>continuous and periodic in frequency</u>

Table 4.1: Comparison of Fourier series and Fourier transform.

https://drive.google.com/file/d/1ov_X0a7tIJzuVRJS1_mJLbIwnpACoeXK/view

Fourier Series (FS)

Basis function
 $e^{jk\omega_0 t}$



Infinite complex
harmonic sinusoids

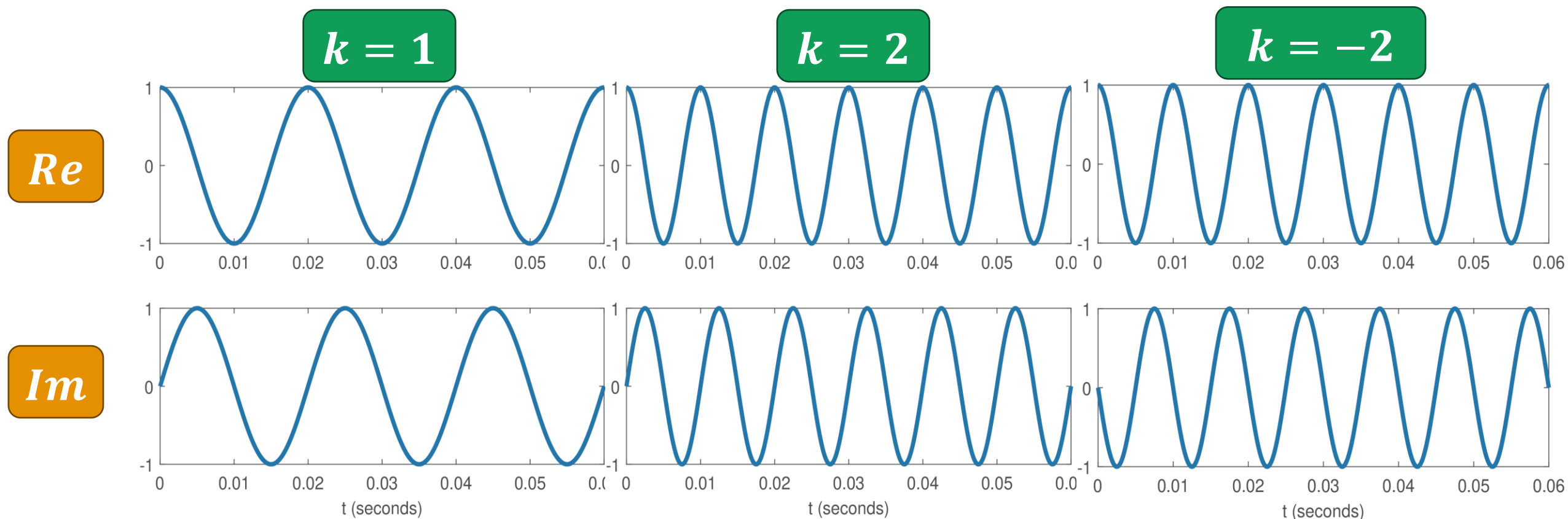
$$c_k = \frac{1}{T_0} \int_{\langle T_0 \rangle} x(t) e^{-jk\omega_0 t} dt$$

$$c_k \Rightarrow \omega = k\omega_0 = k2\pi f_0$$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

$k = 0 \Rightarrow$ DC level
 $k = \pm 1 \Rightarrow$ Fundamental Frequency
 $k = \pm 2, \pm 3 \dots \Rightarrow$ Harmonics

Basis functions of FS



Notice $k = \pm 2$ has same real part but opposite signs at imaginary part, due to even/odd symmetry of sines and cosines

Analyzing a real signal then yields Hermitian symmetry:

$$c_{-k} = c_k^*$$

One-Look Fourier Series (FS)

A sum of sinusoids yields coefficients by inspection

$$\cos(k\omega_0 t) = \frac{e^{jk\omega_0 t} + e^{-jk\omega_0 t}}{2}$$
$$\Rightarrow c_k = c_{-k} = 1/2$$

$$\cos(k\omega_0 t + \phi)$$
$$c_k = 0.5e^{j\phi}$$
$$c_{-k} = 0.5e^{-j\phi}$$

$$x(t) = 3 + \cos(3t) + 7\sin(5t)$$

$$\omega_0 = 1 \text{ rad/s}$$

$$c_0 = 3$$

$$c_3 = 0.5$$

$$c_{-3} = 0.5$$

$$c_5 = 3.5e^{-j\pi/2} = -3.5j$$

$$c_{-5} = 3.5e^{j\pi/2} = 3.5j$$

$$\text{Other } c_k = 0$$

Real signals can also be represented with unilateral spectrum, but the bilateral spectrum is emphasized in the text

Discrete-time Fourier Series (DTFS)

Basis function

$$e^{jk\Omega_0 n}$$

$$\Omega_0 = 2\pi/N_0 \text{ rad}$$



N_0 distinct complex
harmonic sinusoids

Periodic spectrum

$$X[k] = X[k + N_0]$$

$$X[k] = \frac{1}{N_0} \sum_{n=\langle N_0 \rangle} x[n] e^{-jk\Omega_0 n}$$

$$x[n] = \sum_{k=\langle N_0 \rangle} X[k] e^{jk\Omega_0 n}$$

$$x[n] = A \cos\left(\frac{\pi}{5}n + \frac{\pi}{3}\right)$$

$$\Omega_0 = \frac{2\pi}{10}, N_0 = 10$$

$$X[1] = \frac{A}{2} e^{j\frac{\pi}{3}}; X[-1] = \frac{A}{2} e^{-j\frac{\pi}{3}}$$

$$X[0] = X[2] = X[3] = X[4] = X[5] = X[6]$$

$$= X[7] = X[8] = 0$$

$$X[k] = X[k + 10]$$

DFT/FFT to DTFS

When $N = kN_0$

$$DTFS\{x[n]\} = \frac{DFT\{x[n]\}}{N}$$

$k =$ **0** **1** **2** **3** **4** **5**

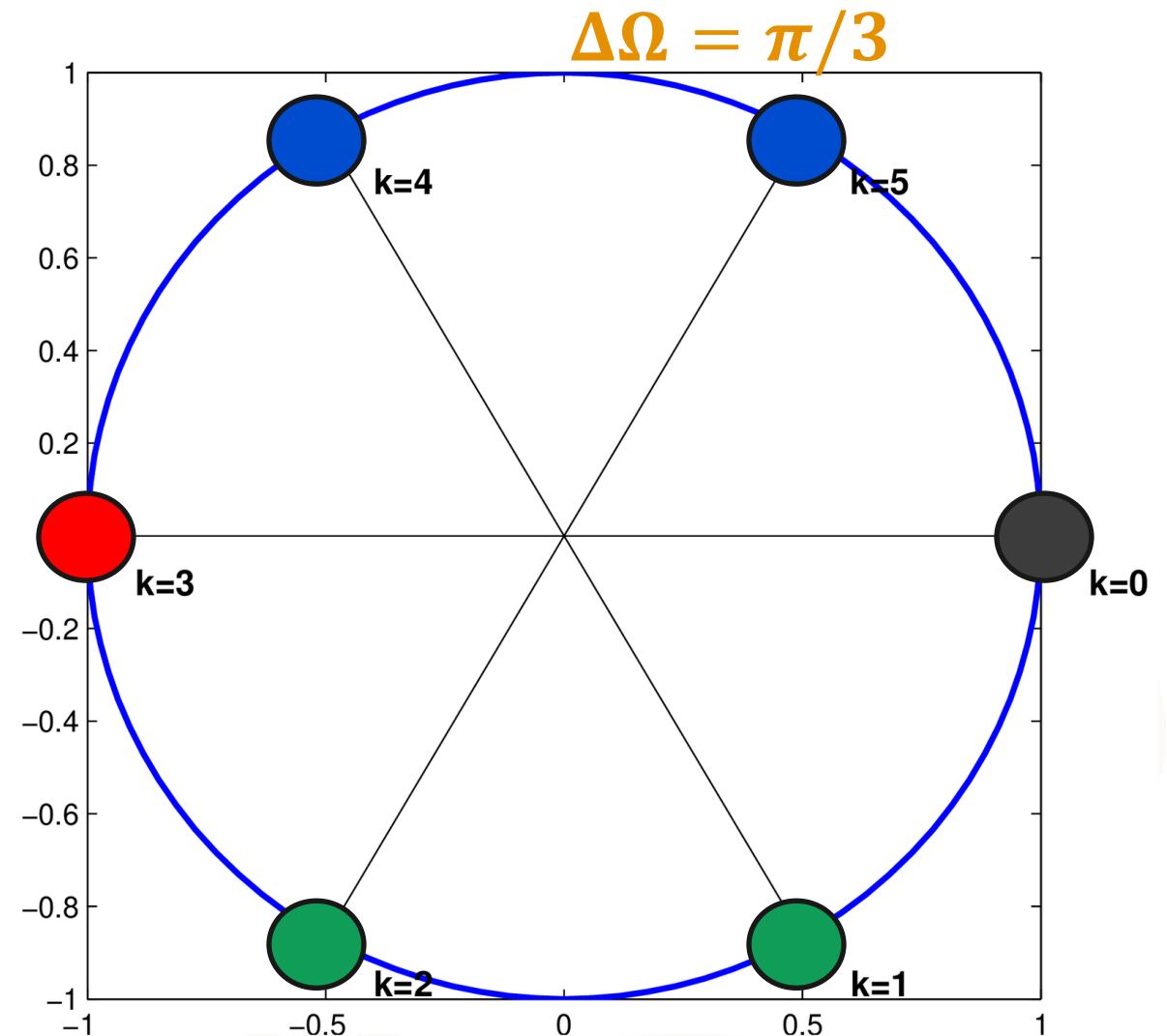
$X[k] =$

0	-1.2j	0.6	0	0.6	1.2j
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$DTFS =$

0	-0.2j	0.1	0	0.1	0.2j
---	-------	-----	---	-----	------

$$x[n] = 0.4\sin\left(\frac{2\pi n}{6}\right) + 0.2\cos\left(\frac{4\pi n}{6}\right)$$



Fourier Transform (FT)

Basis function
 $e^{j2\pi ft}$



Continuum frequency
(V/Hz)

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

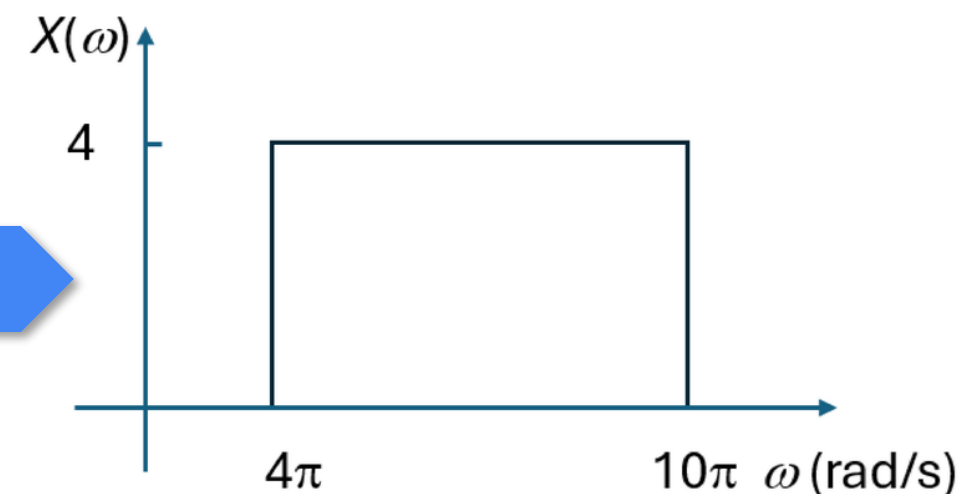
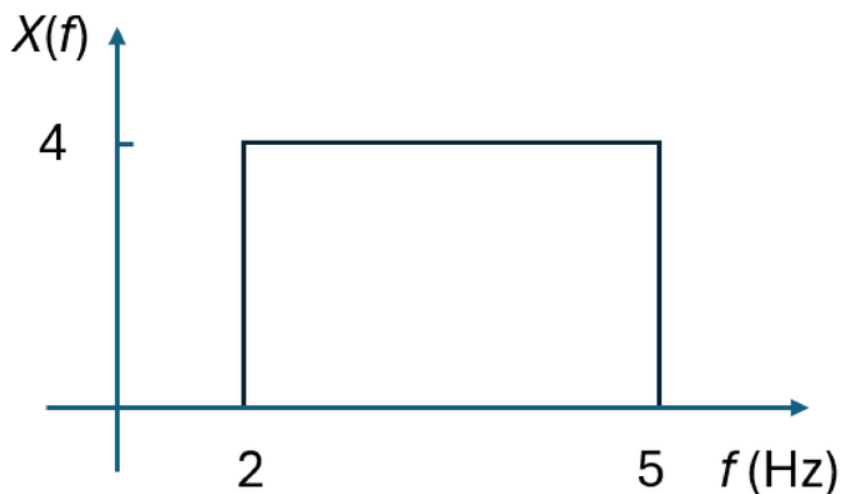
Linear Frequency

Angular Frequency

Fourier Transform (FT) – Linear to Angular Frequencies

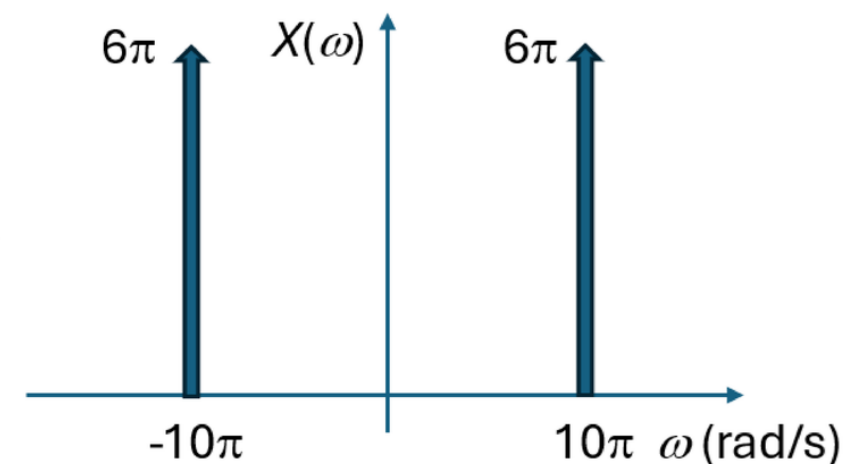
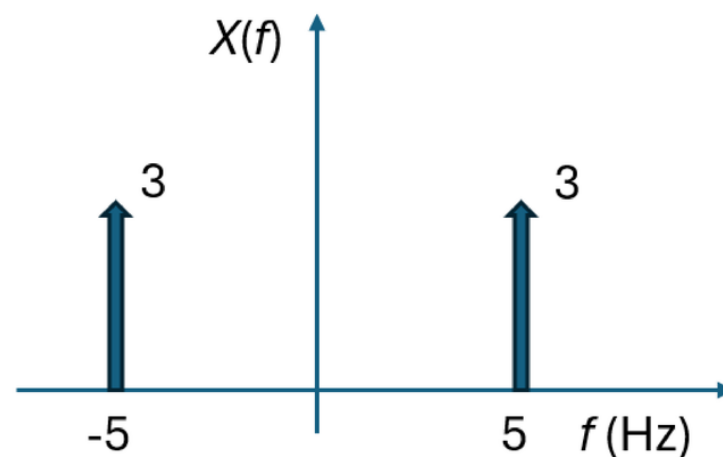
Without
impulses

$$X(\omega) = X\left(\frac{f}{2\pi}\right)$$

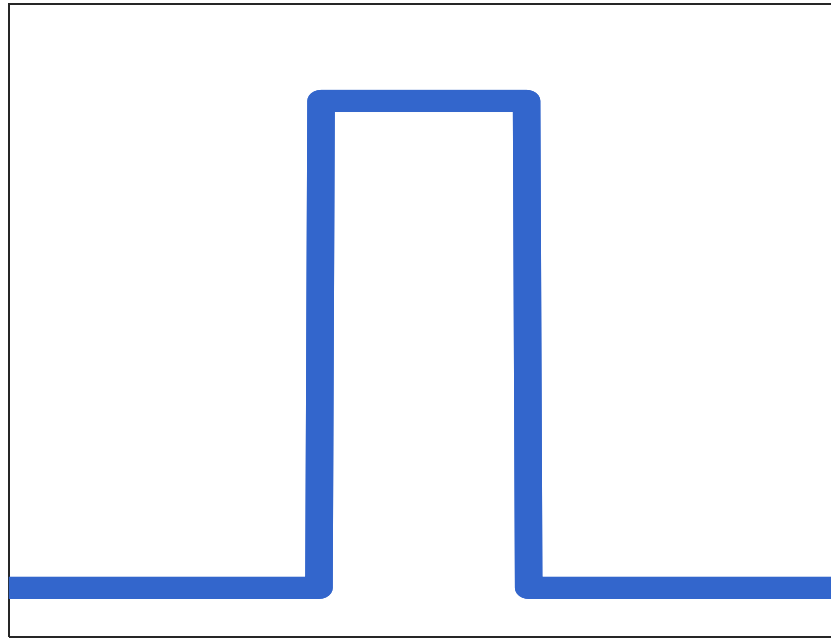


With
impulses

Also scale
impulses by 2π

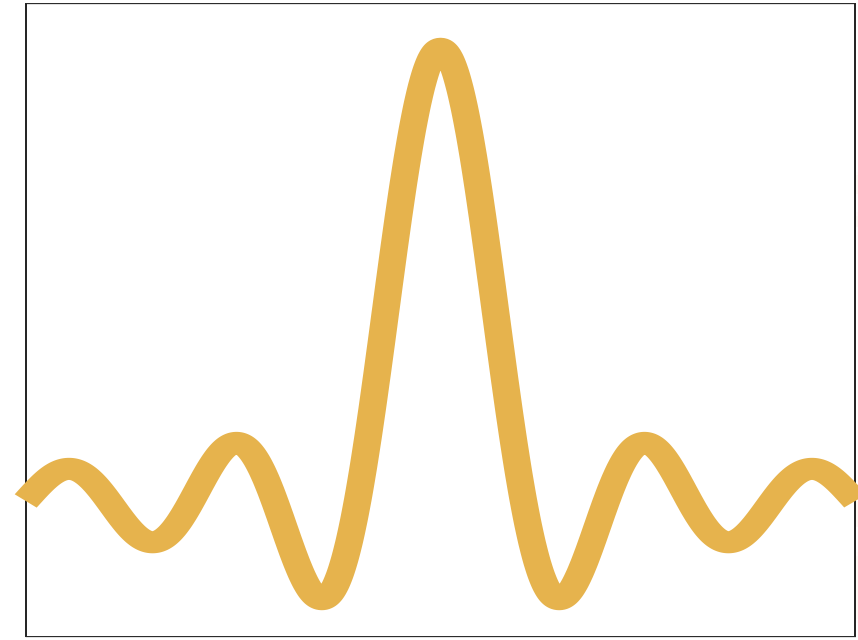


Finite in one domain means infinite in another



rect

finite



sinc

infinite

FT or IFT

FT or IFT

Discrete-time Fourier Transform (DTFT)

Basis function
 $e^{j\Omega n}$



Continuous and
periodic frequency
 (2π) V/rad

$$X(e^{j\Omega}) = \sum_{-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$

Some Properties

$x[n]$

$X(\Omega)$

$x[n - n_0]$

$X(\Omega) e^{-j\Omega n_0}$

$\alpha \delta[n]$

α

DFT/FFT to DTFT

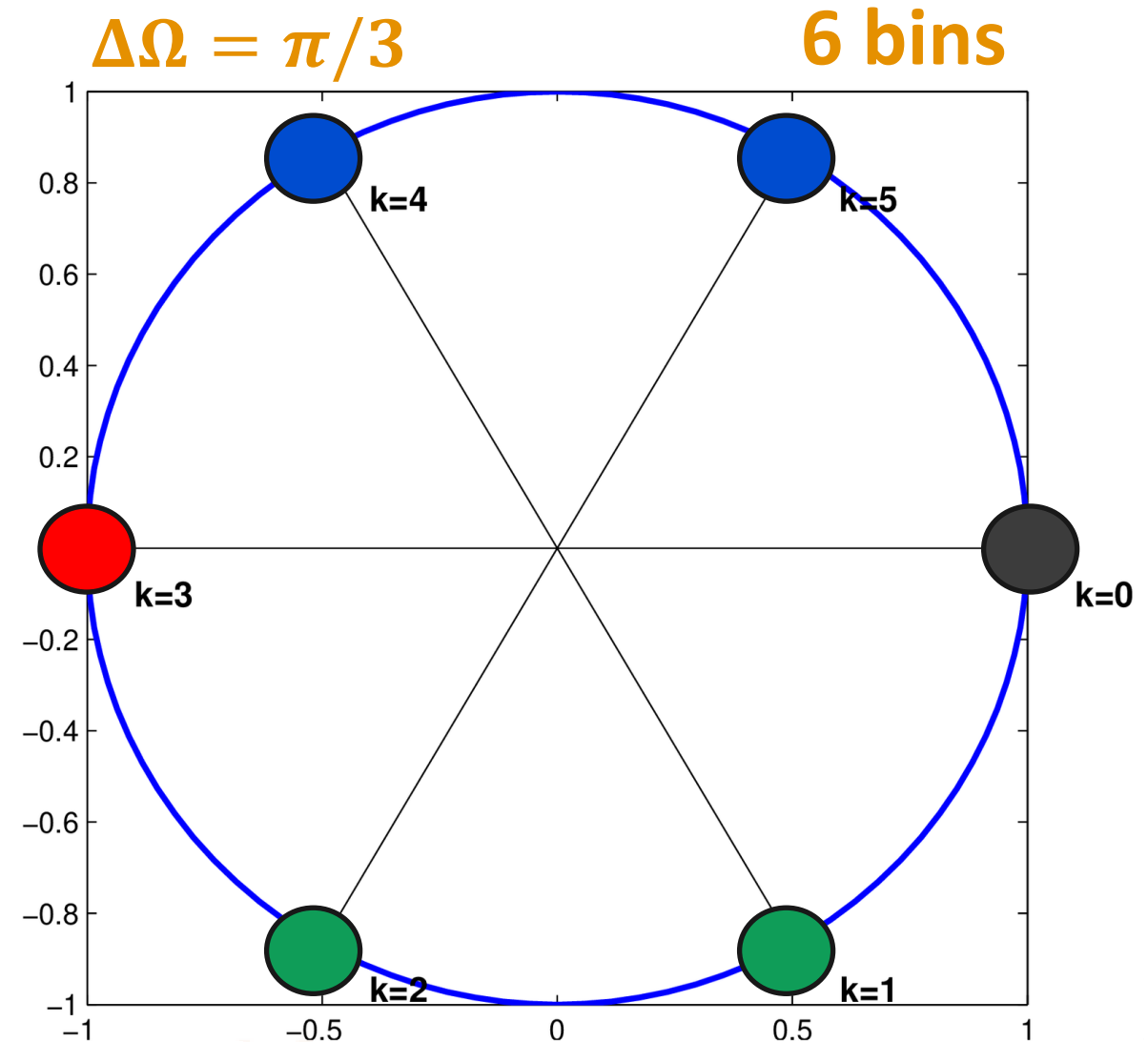
DFT/FFT is a sampling of DTFT

$$X[k] = X(e^{j\Omega}) \Big|_{\Omega = \frac{2\pi}{N}k}$$

$$\Delta\Omega = 2\pi/N$$

$\{0, \Delta\Omega, 2\Delta\Omega, \dots, (N-1)\Delta\Omega\}$ are called DFT bins

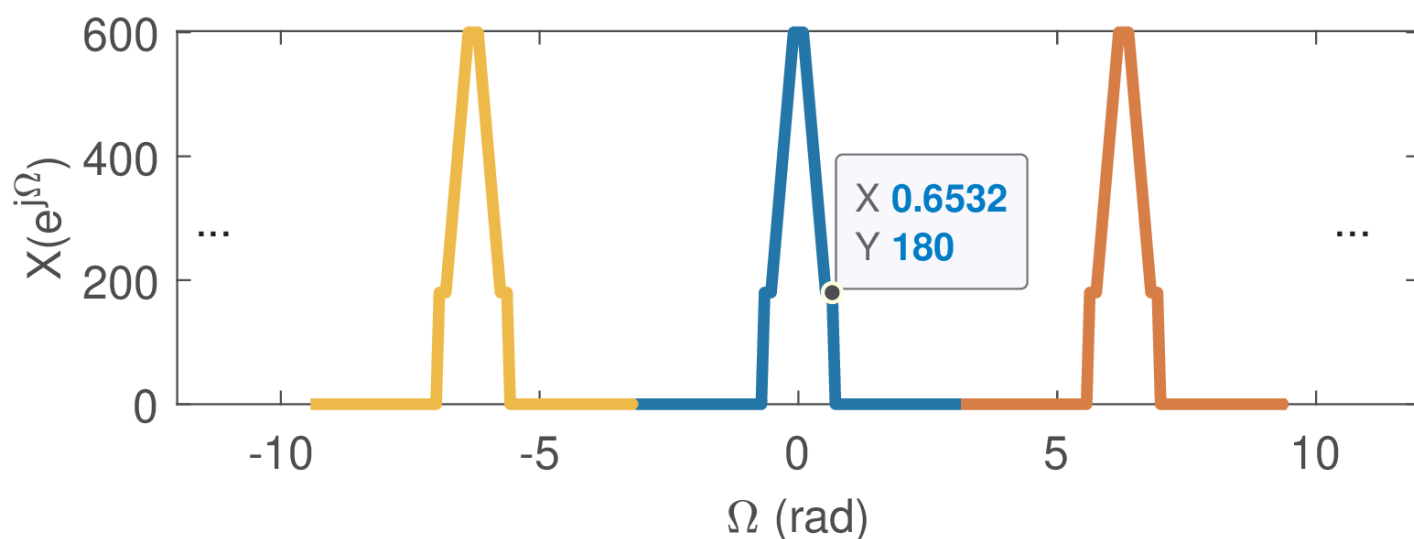
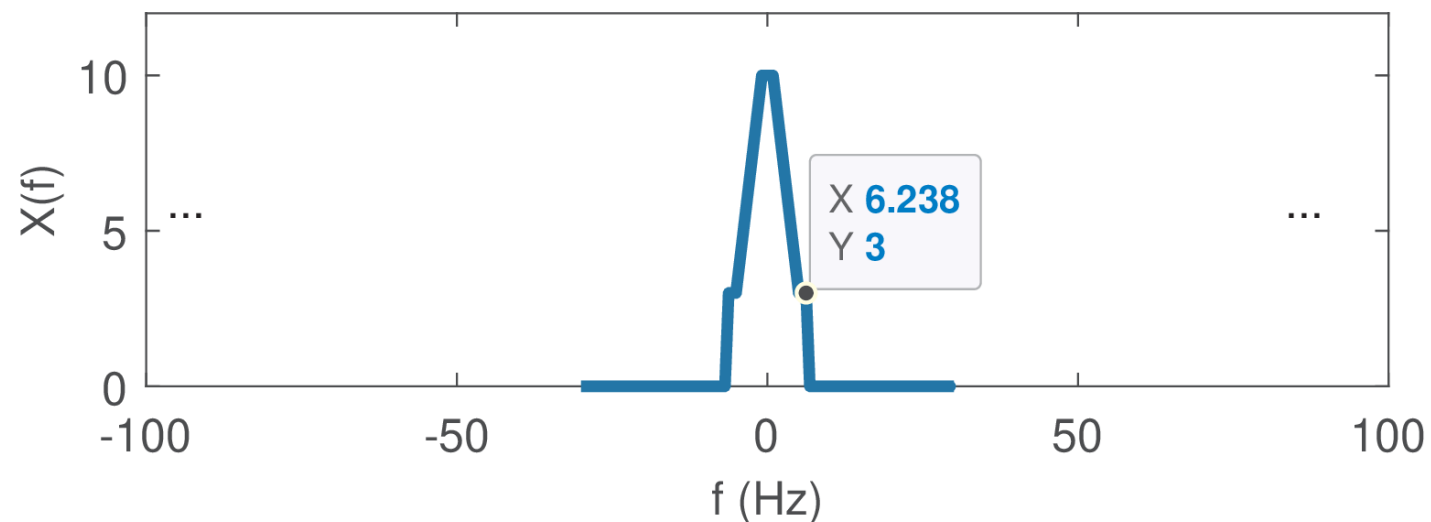
**DTFT → a continuous curve on the unit circle
DFT/FFT → “taking snapshots” of that curve**



AD process generate copies of the original spectrum every 2π

When sampling theorem is obeyed, there is no *aliasing*:

$$X(e^{j\Omega}) = F_s X(\omega) \Big|_{\omega = F_s \Omega}$$



Thank you!

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